

Assignment: HCI Basics

This assignment is designed to test your understanding of user-centered design principles, specifically the Task-Artifact Cycle and task analysis, and your ability to apply these principles to the design of technology.

Objectives:

- To understand the cyclical relationship between tasks and artifacts and explain how this cycle drives the co-evolution of user needs and technology.
- To apply the Task-Artifact Cycle to a real-world example (autonomous driving)
- To identify and define typical user groups for a Smart TV in a family home.
- To identify and prioritize the most common tasks performed on a Smart TV in a family home.

A) Task-Artifact Cycle (10 points)

A.1) Sketch the Task-Artifact Cycle

Sketch the Task-Artifact Cycle as introduced by J. Carroll. Based on the concept, describe how user needs and technologies are evolving together (50 to 100 words).

A.2) Autonomous Driving: Task-Artifact Cycle

Consider the example of autonomous driving in the context of the Task-Artifact Cycle. Discuss, based on the following quote, what needs are articulated and how this impacts the artifacts design. Speculate how this will provide new possibilities and what further human needs may develop (100 to 200 words).

“Human activities implicitly articulate needs, preferences and design visions. Artifacts are designed in response, but inevitably do more than merely respond. Through the course of their adoption and appropriation, new designs provide new possibilities for action and interaction. Ultimately, this activity articulates further human needs, preferences, and design visions.”

B) Users and Tasks: Smart TV (10 points)

B.1) Personas

What are typical users that you would expect for a SmartTV in a family home? What personas would you recommend to create for this use case?

B.2) Tasks

What are typical tasks that you would expect that are carried out on a SmartTV in a family home? Pick the 10 most relevant tasks in your view.

B.3) Task Frequency Analysis

Create a task frequency table where you guess what personas would carry out which tasks and how often they might solve that task (as guessed value from 0.0 to 1.0). Describe what would be your recommendations for the overall design to cater best for all users you imagine (50 to 100 words)?

Submission:

Submit a pdf of your solution for the tasks.

Name your file as a01_id_name.pdf where id is your student id and name is your first name, e.g., a01_R123456_sophie.pdf